

JavaScript Basics

Topics: JavaScript Introduction, let vs const, Data Types, Functions, Arrays & Objects, Array Methods, DOM Basics.

- JavaScript Introduction

JavaScript adds interactivity to web pages. It can read user input, respond to button clicks, modify HTML and CSS, perform calculations, and communicate with servers. JavaScript runs in the browser and can also run on servers using Node.js.

- let vs const

Use **let** for variables whose value will change. Use **const** for variables that should not be reassigned. Although a const object or array cannot be reassigned, its contents can still be modified.

```
let score=0;
score=10;

const student={name:'John'};
student.name='Mary'; // allowed
```

- Data Types

Common data types are Number, String, Boolean, Array, Object, Null and Undefined.

```
let age=15;
let name='John';
let online=true;
let fruits=['Apple', 'Banana'];
```

- Functions

Functions group reusable code.

Traditional:
function add(a,b){ return a+b; }

Arrow Function:
const add=(a,b)=>a+b;
Arrow functions are commonly used in modern JavaScript and React, especially as callbacks for map() and event handlers.

- Arrays & Objects

Arrays store multiple values in order. Objects store related properties using key-value pairs.

```
const colors=['Red', 'Blue'];
const user={name:'Ann', age:16};
```

- Useful Array Methods

push() adds to the end.
pop() removes the last item.
find() returns the first matching item.

findIndex() returns its index.

filter() returns all matching items.

map() creates a new array by transforming each item.

forEach() runs a function for every item.

```
const names=['Ann','Bob'];  
const upper=names.map(n=>n.toUpperCase());
```

- DOM Basics

The DOM (Document Object Model) represents HTML elements as JavaScript objects. Common methods include `getElementById()`, `querySelector()`, `querySelectorAll()`, `innerHTML`, `textContent` and `addEventListener()`.

```
const btn=document.getElementById('btn');  
btn.addEventListener('click',()=>{  
  document.getElementById('title').textContent='Hello';  
});
```